

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 3: Pair of Linear Equations in Two Variables

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Graphical method and consistency

1. In a figure, the graphs of two linear equations are two straight lines that are not parallel and do not meet within the part of the plane that is drawn. The pair of these linear equations is: (a) consistent with unique solution (b) consistent with infinitely many solutions (c) inconsistent (d) inconsistent but can be made consistent by extending these lines **CBSE 2024, 1 mark**

SOLUTION

The two lines are not parallel, so when extended they meet at exactly one point.
A pair of lines that intersect at one point has a unique solution and is consistent.

Answer: (a)

2 Elimination method

2. Solve the following system of linear equations $7x - 2y = 5$ and $8x + 7y = 15$ and verify your answer. **CBSE 2024, 2 marks**

SOLUTION

Multiply the first equation by 7 and the second by 2: $49x - 14y = 35$ and $16x + 14y = 30$

Add: $65x = 65$, so $x = 1$. From $7x - 2y = 5$: $7 - 2y = 5$, so $y = 1$.

Verification: $7(1) - 2(1) = 5$ and $8(1) + 7(1) = 15$. Both are true.

Answer: $x = 1, y = 1$

3 Word problems

3. Three years ago, Rashmi was thrice as old as Nazma. Ten years later, Rashmi will be twice as old as Nazma. How old are Rashmi and Nazma now? **CBSE 2024, 3 marks**

SOLUTION

Let Rashmi be x years old and Nazma y years old now.

Three years ago: $x - 3 = 3(y - 3)$, so $x = 3y - 6$.

Ten years later: $x + 10 = 2(y + 10)$, so $x = 2y + 10$.

$3y - 6 = 2y + 10$ gives $y = 16$, and $x = 42$.

Answer: Rashmi is 42 years old and Nazma is 16 years old.